3.) t memenuhi persamaan $s(t) = 3t - 24t^2$ untuk $t \geq 0$

a. Hitung kecepatan rata-rata antara $t = 1{,}50$ sec , $t = 1{,}54$ sec , berapa perbedaannya

$\Delta t = 1{,}54 - 1{,}5 = 0{,}04$

$V = \frac{ds}{dt} = d(3t - 24t^2) = 3 - 48t$

$V = \frac{\Delta s}{\Delta t} = \frac{V(1{,}54) - V(1{,}5)}{\Delta t} = \frac{(3-48)(1{,}54) - (3-48)(1{,}5)}{0{,}04}$

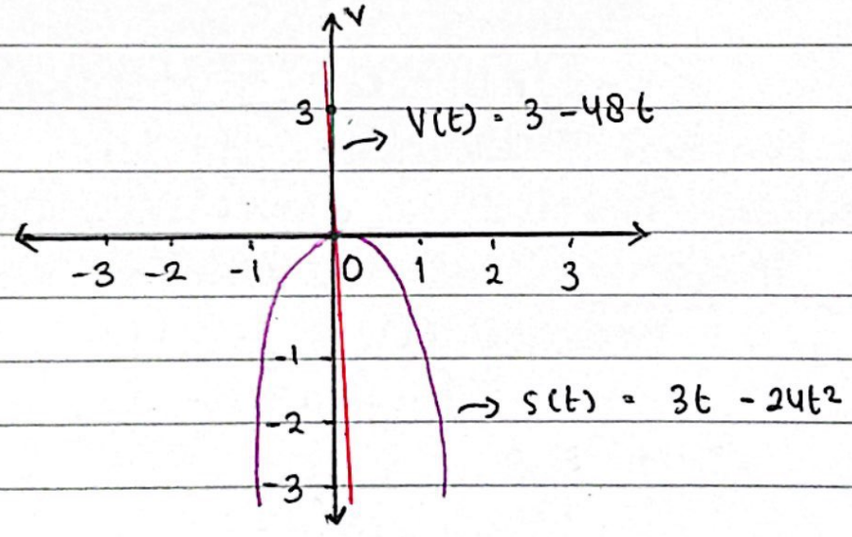
$V = \frac{-70{,}92 + 69}{0{,}04} = -48$ m/s //

$t = 1{,}5$

$V(1{,}5) \rightarrow V = (3-48)(1{,}5) = -69$ m/s

∴ Jadi perbedaannya = 21 m/s

6. Grafik Fungsi S(t) dan V(t)



Dipindai dengan CamScanner